Criticality in self-organized percolation

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Self-organized percolation reaches the critical point of a lattice with no tuning, through a global feedback on the occupation probability p, but the origin of that feedback and the reason it stops at the threshold have remained unclear. We show that the feedback rule follows from the volume balance of fluid injected at a constant rate into a compressible system, so that its target measures the flow and its strength measures the stiffness. The plateau of p is then the value at which the stationary front density at fixed p equals the target: the feedback selects a state that exists without it, and the plateau approaches the threshold because that density vanishes only there. A scaling argument gives the density an exponent $x = \beta + \nu(d_{\min} - 1) \approx 0.313$ with no fitted parameters, in agreement with simulations of site and bond percolation, and therefore a relation between the plateau and the rate. This allows the threshold to be located from plateaus at several rates, without tuning the rate to the size of the system, and it shows that the rate sets the scale up to which the cluster is fractal. The plateau does not depend on the initial probability, on the stiffness at moderate feedback or, beyond a boundary layer, on the layout of the seeds. We also discuss why the one-chance growth rule of the model prevents this reading from describing slow drainage.

1. Introduction

Percolation provides the simplest description of connectivity in a disordered medium [1,2], and flow through rocks has been among its most natural applications [3]. Standard percolation is nevertheless a static construction: the occupation probability p is an external parameter, and criticality is observed only when it is tuned to the threshold p_c. The growth algorithm of Leath [4] keeps this limitation, since a critical cluster is grown only if p is fixed at p_c in advance. Nature rarely offers such tuning, and the question of how disordered systems reach critical states without it lies at the heart of self-organized criticality [5].

Alencar, Andrade and Lucena answered this question for cluster growth with self-organized percolation (SOP) [6]. They let the occupation probability vary during Leath growth, through a simple global rule that regulates the number of active sites in the growth front. On the square lattice, p then reaches a stationary state with $\langle p \rangle \approx 0.59$ for site percolation and $\langle p \rangle \approx 0.5$ for bond percolation, in close agreement with the known thresholds and independently of its initial value. No parameter is tuned, and the same behavior is observed for different feedback parameters and even when the regulated quantity grows in time as a power law. The system thus remains close to p_c while the cluster changes from a compact object into a ramified structure resembling a percolation cluster, which also allows p_c to be estimated in a single run. Later work extended the behavior to other regular lattices and to disordered networks [7].

These results left the mechanism open. First, the feedback was introduced as a controlling rule, and its physical origin was not identified: the original work pointed only to branched polymer

growth and tumor vascular networks as possible realizations [6]. Second, and for the same reason, the strength of the feedback and its target have no measurable counterparts. Third, and most important, the plateau was observed but not explained: nothing in the rule mentions p_c, so it is not obvious why p should stop there, or how far from it the plateau lies at a given target.

Here we argue that the SOP feedback has a direct physical realization in the injection of a fluid into a porous medium at a constant rate. This is a setting in which percolation ideas are already well established. When a non-wetting fluid slowly displaces a wetting one, capillary forces dominate and the invader advances through the throat with the lowest entry pressure. This is the invasion percolation of Wilkinson and Willemsen, which finds the critical point of the medium automatically [8]. Experiments and network simulations place this capillary regime within a phase diagram spanned by the capillary number and the viscosity ratio [9], and viscous forces modify the invasion pattern as the rate increases [10].

Invasion percolation, however, assigns no role to the injection rate or to the elastic response of the system, and it therefore says nothing by itself about the pressure, which is the quantity most readily measured. Under constant-rate drainage the pressure is observed to jump abruptly [11]. Hansen and coworkers have followed its temporal evolution in network simulations that resolve capillary and viscous forces [12], and have identified bursts of invasion with pressure drops, comparing simulated signals with experiments at low and high injection rates [13]. A similar intermittency arises when throats open in suspension flows. Herrmann and collaborators measured power-law distributed bursts in the pressure drop of a particle-laden flow through a bed of glass beads, and suggested a new experimental realization of self-organized criticality [14]. Simulations then attributed such bursts to clogged pores that reopen once the pressure gradient across a deposit exceeds a threshold, recovering the experimental distribution of pressure-loss jumps [15]. Together with numerical work by Andrade, Herrmann and coworkers showing how pore geometry, rheology and inertia combine to shape flow in disordered media [16], these results suggest a common picture: a pressure shared across the medium opens pathways, and the opening of those pathways in turn relaxes the pressure.

SOP expresses this picture in a minimal form, and here we build the connection explicitly. Starting from the capillary entry rule and the volume balance of a compressible injection system, we recover the feedback rule of Alencar, Andrade and Lucena, which answers the first two questions: the target of the feedback measures the imposed flow per unit area of the inlet wall, and its strength measures the stiffness of the injection system. Keeping the growth rule of SOP, in which each pore has one chance to be filled, we then answer the third. The plateau of p is the value at which the front density at fixed p equals the target, which gives a relation between the plateau and the rate, with an exponent set by percolation. We use it to locate the threshold at fixed rate, and to show how the rate limits the fractal range of the cluster. We close by discussing how far this reading goes, and why the one-chance rule limits it as a description of real flow.

2. Model

We consider the injection of a non-wetting fluid, at a constant volumetric rate Q, through one wall of a porous medium initially filled with a wetting fluid. The medium is a network of pores connected by throats, with transverse size L in d dimensions, and $A = L^{d-1}$ is the wall surface area in lattice units. The fluid enters through seed pores on the wall. We work in the capillary regime, where viscous pressure drops are negligible and the invading fluid has a single pressure $P(t)$. As a simplification, we give each pore a single effective radius that controls its entry, whichever throat the fluid arrives through.

A pore of effective radius r_i can be invaded only when the pressure exceeds its entry pressure, which by the Young–Laplace law is

$$P_i = \frac{2\gamma \cos\theta}{r_i}, \tag{1}$$

where γ is the interfacial tension and θ is the contact angle. Disorder in r_i makes the entry pressures independent random numbers with a common distribution. Let $\tilde{P}$ be the entry pressure of a pore chosen at random, with cumulative distribution $F(P) = \Pr\big(\tilde{P} < P\big)$, whose derivative $F' = \mathrm{d}F/\,\mathrm{d}P$ is positive. We define the rank $x_i = F(P_i)$ of pore i, which is the fraction of pores with a lower entry pressure. Because F is continuous and increasing, for a pore chosen at random,

$$\Pr\big[F\big(\tilde{P}\big) < x\big] = \Pr\big[\tilde{P} < F^{-1}(x)\big] = F\big[F^{-1}(x)\big] = x, \tag{2}$$

so the x_i are uniform on $[0, 1]$ for any distribution of entry pressures. The same property of F gives

$$P_i < P \quad \Longleftrightarrow \quad x_i < F(P). \tag{3}$$

The capillary criterion therefore takes the form of a percolation test, $x_i < p$, with the occupation probability

$$p(t) = F[P(t)], \tag{4}$$

which is the fraction of pores that the current pressure is able to enter.

This fixes the growth rule. The perimeter of the invading cluster is the set of empty pores next to it. A pore is tested once, at the step in which it is first reached: if $x_i < p(t)$ it is filled, and otherwise it stays empty for the rest of the growth. The $N(t)$ pores filled at time t are the active sites of the growth front, whose empty neighbors are tested at the next step. This one-chance rule is a simplification, since a real pore could be entered later if the pressure rose above its threshold, and we adopt it throughout this work.

The pressure follows from volume conservation. Part of the injected fluid fills pores, at a rate $q(t)$, and the rest is stored by raising the pressure of the system. Defining the compliance C as the volume stored per unit increase of pressure, which includes the compressibility of the fluid and of the injection line, we have

$$C\frac{\mathrm{d}P}{\mathrm{d}t} = Q - q(t). \tag{5}$$

Measuring the front in units of the wall surface area, with $f(t) = N(t)/A$, and taking pores of volume v that each fill in a time τ,

$$q(t) = \frac{f(t)Av}{\tau}. \tag{6}$$

We take τ as the time step of the growth, so that the front advances one generation of pores per step. Integrating Eq. (5) over one step, with f constant during the step, gives

$$C[P(t+\tau) - P(t)] = Q\tau - f(t)Av, \tag{7}$$

and factoring out Av gives

$$P(t+\tau) - P(t) = \frac{vA}{C}\left[\frac{Q\tau}{vA} - f(t)\right]. \tag{8}$$

The term in brackets compares the pores that the imposed flow would fill in one step with those actually filled. Applying Eq. (4) and expanding F around the stationary pressure P^* gives

$$p(t+1) - p(t) = F'(P^*)[P(t+\tau) - P(t)] + \mathcal{O}(\Delta P^2), \tag{9}$$

where $t+1$ now counts steps of length τ and ΔP is the pressure change in one step. Together with Eq. (8), this yields a closed rule for the occupation probability,

$$p(t+1) = p(t) + c[f_T - f(t)], \tag{10}$$

with

$$f_T = \frac{Q\tau}{vA}, \qquad c = \frac{vAF'(P^*)}{C}. \tag{11}$$

The target fraction f_T is the number of pores per unit wall area that the imposed flow would fill in one step, so it measures the injection rate. Averaging Eq. (8) over a stationary state, in which the pressure does not drift, gives $\langle f \rangle = f_T$, or $\langle q \rangle = Q$: on average the pores take up exactly the injected flow, and c does not enter this balance. The feedback strength c is the change in p caused by filling one layer of A pores, so it measures the stiffness of the system: a stiff system has a large c and a soft one a small c. When the compliance comes from a volume V_c of fluid with compressibility β, so that $C = \beta V_c$, it reads $c = (vA/V_c)F'(P^*)/\beta$, proportional to the ratio between the pore volume of one layer and the compressible volume. The two parameters can be set independently, through Q and C.

The result is exact for all P if the thresholds are uniformly distributed over a width W, where $F' = 1/W$ and $c = vA/(CW)$. For other distributions it holds when ΔP is small compared with the scale over which F' varies.

For a pore of size a, the capillary pressure γ/a drives a fluid of viscosity μ at a velocity of order γ/μ, so the filling time is $\tau \sim \mu a/\gamma$. With $v \sim a^d$ and an injection velocity $u = Q/\left(Aa^{d-1}\right)$, the first relation in Eq. (11) becomes

$$f_T \sim \frac{\mu u}{\gamma} = \mathrm{Ca}, \tag{12}$$

up to a geometric factor, where Ca is the capillary number. Small f_T is thus the capillary limit of drainage, which is also where a single pressure $P(t)$ is justified.

Eq. (10) is the feedback rule of self-organized percolation. In the original notation it reads $p(t+1) = p(t) + k[N_T - N(t)]$, where k is a kinetic coefficient and N_T is a target number of active sites [6], and our derivation identifies $N_T = f_T A = Q\tau/v$ and $k = c/A = vF'(P^*)/C$. The growth rule is also that of SOP, where, as in the Leath algorithm [4], a perimeter site that fails its test remains unoccupied forever [6]. Neighboring filled pores are connected, so a cluster is a set of connected filled pores, as in site percolation. Placing the thresholds on the throats instead gives bond percolation, which we also consider, and which has the same large-scale behavior. A complete run is shown in Fig. 1, and the first steps of the growth are illustrated in Fig. S1 of the Supplemental Material.

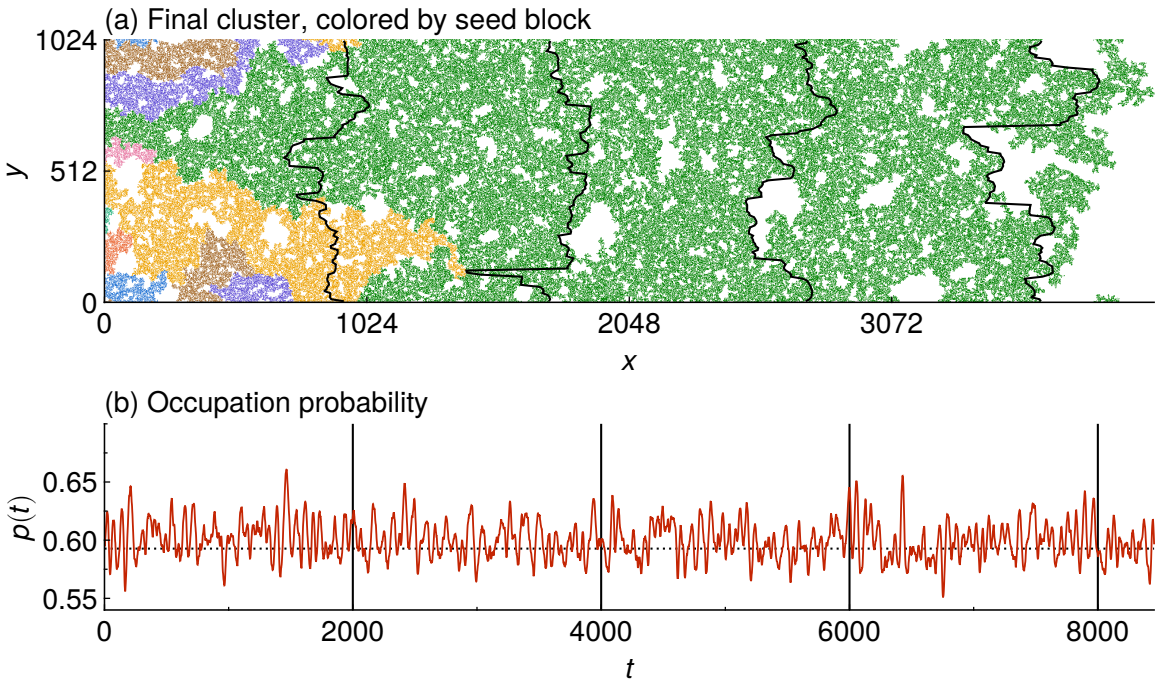


Figure 1: One complete run, for site percolation with $L = 2^{10}$, a lattice of length $4L$, $c = 0.05$, $f_T = 0.2$ and $p(0) = 0.6$, with periodic sides. The seeds form 8 evenly spaced blocks of 64 sites on the left wall, so the seed density on the wall is $\rho_0 = 1/2$. (a) Final cluster, with pores colored by the seed block they grew from, traced through neighbors filled one step earlier. Black lines are frontlines, the farthest x reached at $t = 2000, 4000, 6000$ and 8000, in blocks of 8 sites along y. (b) Occupation probability of the same run, with vertical lines at those times. The dotted line marks p_c.

The derivation assumes that the pressure is uniform across the cluster, that all pores have the same volume v, that they fill synchronously in a time τ, and that C is constant. Since p stays close to p_c, the relevant pressure is $P^* \approx F^{-1}(p_c)$. The restriction of p to $[0, 1]$ corresponds to pressures outside the range of the thresholds, where F saturates at 0 or 1. This correspondence is not exact, because clipping p discards the excess while a real pressure keeps building, but it should only matter in early transients. Viscous pressure gradients, trapping of the wetting fluid and gravity are neglected.

3. Self-organized injection pressure

Fig. 2 shows that the injection pressure forgets where it started. For very different initial values of p, the curves reach the same plateau, close to p_c, after a short transient. The number of inlet pores matters even less, as shown in Fig. S2 of the Supplemental Material. Fig. 2 also varies the stiffness of the system through c. A soft system, with small c, responds slowly, with a smaller overshoot and a longer transient, and a stiff one responds quickly, with a larger overshoot. In both cases p reaches the same plateau, as expected from the stationary balance $\langle f \rangle = f_T$, in which c does not appear.

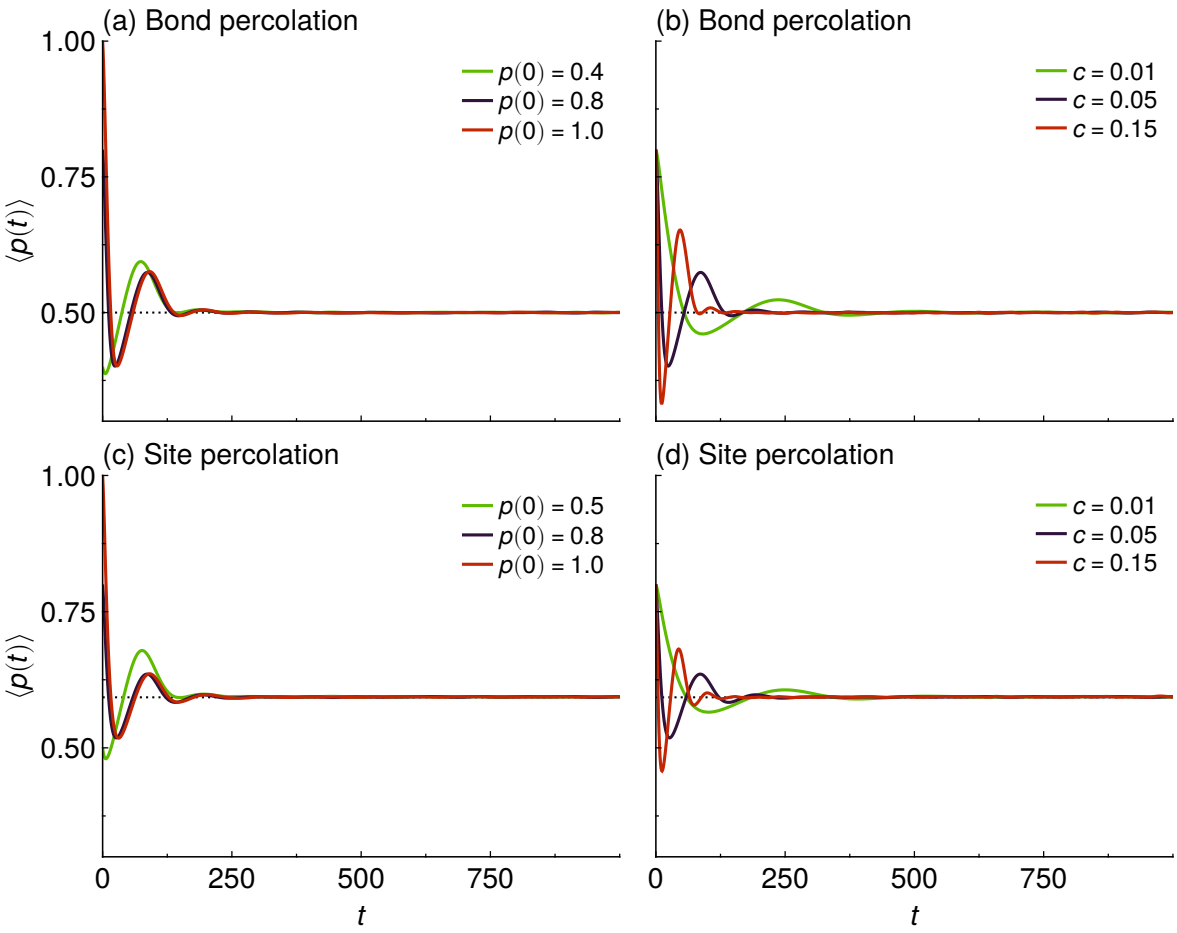


Figure 2: Independence of the initial pressure and of the stiffness. Mean $p(t)$ for bond and site percolation, with $L = 2^{14}$, $f_T = 0.1$ and a seed density $\rho_0 = 0.2$ on the wall, with seeds placed at random on the wall. (a, c) Different $p(0)$ with $c = 0.05$. (b, d) Different c with $p(0) = 0.8$. Each curve is an average over 200 runs, none of which went extinct. Dotted lines mark p_c.

Uses: Fig. 3 of Germano's version.
Missing: Fig. 3 extended to the plateau for $L = 16384$; correct f_T in the text for Fig. 3; review of the claim that larger c settles closer to criticality.

4. Pressure–rate curve

The occupation probability is directly related to the percolation threshold of the medium. At a fixed pressure P, mark every pore with $P_i < P$ as able to be entered. Each pore is marked with probability $F(P)$, independently of the others because the entry pressures are independent, which is exactly site percolation with occupation probability $p = F(P)$. The marked pores therefore first connect across the medium at the critical pressure

$$P_c = F^{-1}(p_c). \tag{13}$$

Thresholds on throats give the same relation with the bond value of p_c. Note that p counts all pores the pressure could enter, anywhere in the medium, whereas the invaded cluster contains only those connected to the inlet. Growth at a fixed p gives exactly a percolation cluster, since each pore is tested once with the same probability. When p varies in time this is only an approximation, because different pores are tested at different values of p. It is justified as long as p stays near a single value, and the comparison below tests it.

Why does the plateau sit near p_c? The answer does not need the feedback. Consider growth from the wall at a fixed p, with $c = 0$. Below p_c clusters are finite and the front dies, while above p_c it survives with a stationary density $f_\infty(p) > 0$. Since $\langle f \rangle = f_T$ in the stationary state, the feedback only has to find the value of p at which this density equals the target, so

$$p^* = f_\infty^{-1}(f_T). \tag{14}$$

As f_∞ vanishes only at p_c, a thin front can be sustained only near the threshold, and p^* approaches p_c as f_T decreases. In the language of injection, a slow flow can only be carried at the pressure where the medium barely conducts.

Panels (b) and (c) of Fig. 3 test Eq. (14). We ran SOP at a given f_T to obtain p^*, and then fixed p at that value, with no feedback, and measured the density that the front reaches. At $L = 2^{14}$, the ratio $f_\infty(p^*)/f_T$ lies between 1.00 and 1.04 for $f_T \geq 0.2$ in both bond and site percolation, and the points follow the curve up to $f_T = 0.8$, the highest rate tested. The two systems therefore sit at the same point, and the feedback selects a state that already exists without it.

The form of f_∞ follows from the geometry of the cluster. A front that advances at speed v and leaves behind a cluster of density P_∞ has $f_\infty \approx v P_\infty$. Near p_c, $P_\infty \sim (p - p_c)^\beta$ and, since a distance ξ takes $\xi^{d_{\min}}$ steps to cross, $v \sim \xi^{1-d_{\min}} \sim (p - p_c)^{\nu(d_{\min}-1)}$. Hence

$$f_\infty \sim (p - p_c)^x, \qquad x = \beta + \nu(d_{\min} - 1) \approx 0.313, \tag{15}$$

with $\beta = 5/36$, $\nu = 4/3$ [2] and $d_{\min} \approx 1.1308$ [17]. The measured local exponent lies between 0.29 and 0.32 for $0.007 \leq p - p_c \leq 0.02$ in both cases, with no fitted parameters in Eq. (15). Together, Eq. (14) and Eq. (15) give the pressure–rate curve,

$$p^* - p_c \sim f_T^{1/x} \approx f_T^{3.2}, \tag{16}$$

so that, through Eq. (12), the plateau should approach its critical value as $\mathrm{Ca}^{3.2}$.

Fig. 3(a) shows this curve directly, with only the injection rate changing: L, c, $p(0)$ and ρ_0 are fixed. At low rates p^* stays close to p_c, and the distance from it rises smoothly with f_T. Between $f_T = 0.2$ and $f_T = 0.5$, the local exponent of $p^* - p_c$ against f_T lies between 2.9 and 3.2 for site percolation and between 2.7 and 3.3 for bond percolation, in agreement with Eq. (16). At higher rates the system is far from p_c and Eq. (16) no longer applies. At lower rates the curve flattens, with $p^* - p_c$ of the order of $L^{-1/\nu}$, which suggests a finite-size limit on how close to p_c the plateau can get at fixed L.

At small f_T the SOP points leave the curve: at $f_T = 0.1$ and $L = 2^{14}$ the ratio $f_\infty(p^*)/f_T$ is 1.46 for bond and 1.15 for site percolation. The gap shrinks from $L = 2^{12}$ to $L = 2^{14}$. There the correlation length is comparable to L, the relaxation of the fixed-p runs may be incomplete, and the fluctuations of p are largest, and we have not separated these effects.

Uses: Fig. 6 of Germano's version; the c panels of Fig. 6 of Junior's version, if restored.
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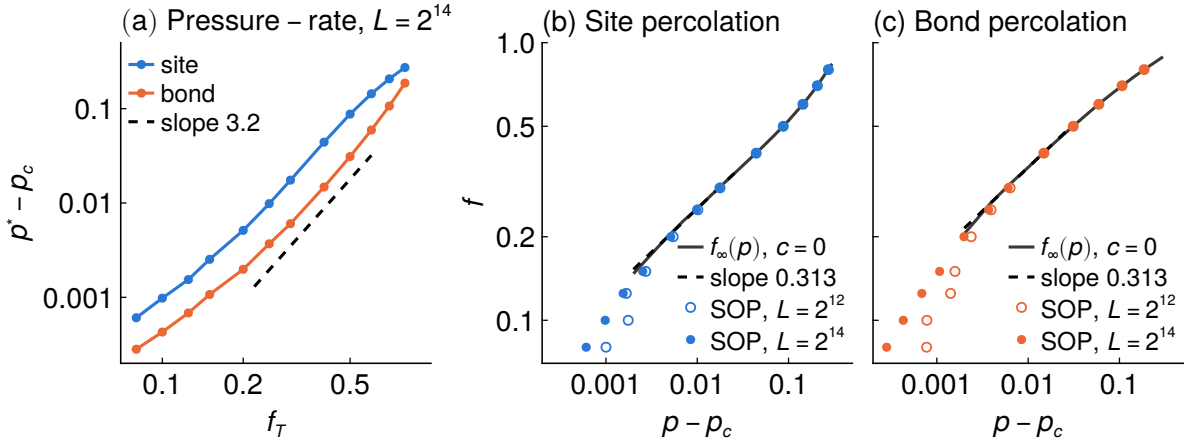


Figure 3: The plateau and the front density. (a) Distance of the plateau from p_c against the injection rate, with $L = 2^{14}$, $c = 0.05$, $\rho_0 = 0.2$ and random seeds, averaged over 32 runs to $t = 8000$ and the second half of each run. The dashed line has the slope of Eq. (16). (b, c) Dark lines show f_∞ at fixed p with $c = 0$, for $L = 2^{14}$, starting from a full wall, with 8 runs to $t = 16000$. Symbols show $(p^* - p_c, f_T)$ from the runs of (a), at $L = 2^{12}$ and $L = 2^{14}$. Dashed lines have the slope of Eq. (15).

5. Critical pressure from rate extrapolation

Eq. (16) suggests a way of finding the critical pressure of a medium by varying only the injection rate, and it needs no tuning of the rate with the size of the system. We proceed in two steps, shown in Fig. 4. At fixed f_T, the plateau decreases with L, and we fit $p^*(L) = p_\infty(f_T) + aL^{-1/\nu}$ for $L \geq 2^{11}$, with $\nu = 4/3$ taken from percolation and not measured here. The limits p_∞ then fall on a straight line against $f_T^{1/x}$, as Eq. (16) predicts, and its value at $f_T = 0$ is the estimate of p_c.

For $f_T \leq 0.4$ this gives $p_c = 0.59270(4)$ for site percolation and $p_c = 0.50020(4)$ for bond percolation, against the known values 0.59275 and 0.5. These errors are statistical, and the result also depends on the range of the fit: for $f_T \leq 0.3$ we find 0.59295(6) and 0.50024(6), and for $f_T \leq 0.25$, 0.59306(8) and 0.50019(8). The method therefore locates p_c to about 3×10^{-4}. Through Eq. (13), this is a measurement of the critical pressure P_c of a medium from plateau pressures at several injection rates.

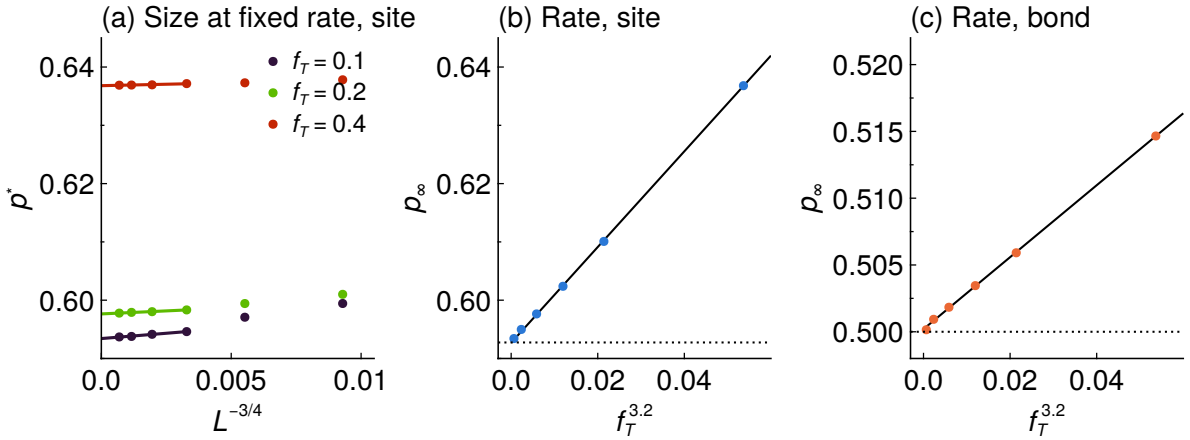


Figure 4: Extrapolation at fixed rate, with $c = 0.05$, $\rho_0 = 0.2$ and random seeds. (a) Plateau against $L^{-3/4}$ for three values of f_T, in site percolation, with fits for $L \geq 2^{11}$, from $L = 2^9$ to $L = 2^{14}$. (b, c) Extrapolated plateau p_∞ against $f_T^{1/x}$ for site and bond percolation, with fits for $f_T \leq 0.4$. Dotted lines mark p_c. Error bars are smaller than the symbols.

6. Invasion pattern

A fixed rate also gives a simple picture of the invasion pattern. As f_T sets the distance $p^* - p_c$, it sets a correlation length $\xi \sim (p^* - p_c)^{-\nu}$. Below ξ the cluster should look like a critical percolation cluster, and above it like a uniform medium with holes. To measure this, we take windows of side L along the strip, past the boundary layer near the wall, as in Fig. 5(a). Inside each window, using only its sites, we keep the largest cluster that connects the two ends of the window along the flow. We then count the boxes of side r that contain it, $N(r)$, and measure the shortest path inside it from the start of the window to a distance r, $D(r)$.

Fig. 5(b–d) shows the result for site percolation at $L = 2^{14}$. At the lowest rate, $f_T = 0.1$, ξ is about the size of the window, and the local slope of $N(r)$ stays between 1.89 and 1.92 from $r = 16$ to $r = 2048$, close to the percolation value $d_f = 91/48 \approx 1.896$. Its local path dimension stays between 1.11 and 1.18, around $d_{\min} \approx 1.131$. At higher rates both slopes leave these values

and reach the bulk values 2 and 1 at a scale that decreases with f_T, and they do so when r is a fraction of ξ. The fractal pattern is therefore limited by the injection rate: a slow injection leaves a fractal cluster over a wide range of scales, and a fast one is fractal only at short distances. The density of the cluster shows the same trend, rising from 0.26 at $f_T = 0.1$ to 0.59 at $f_T = 0.4$.

These are local slopes from 12 windows, which show the crossover but are not precise estimates of the exponents.

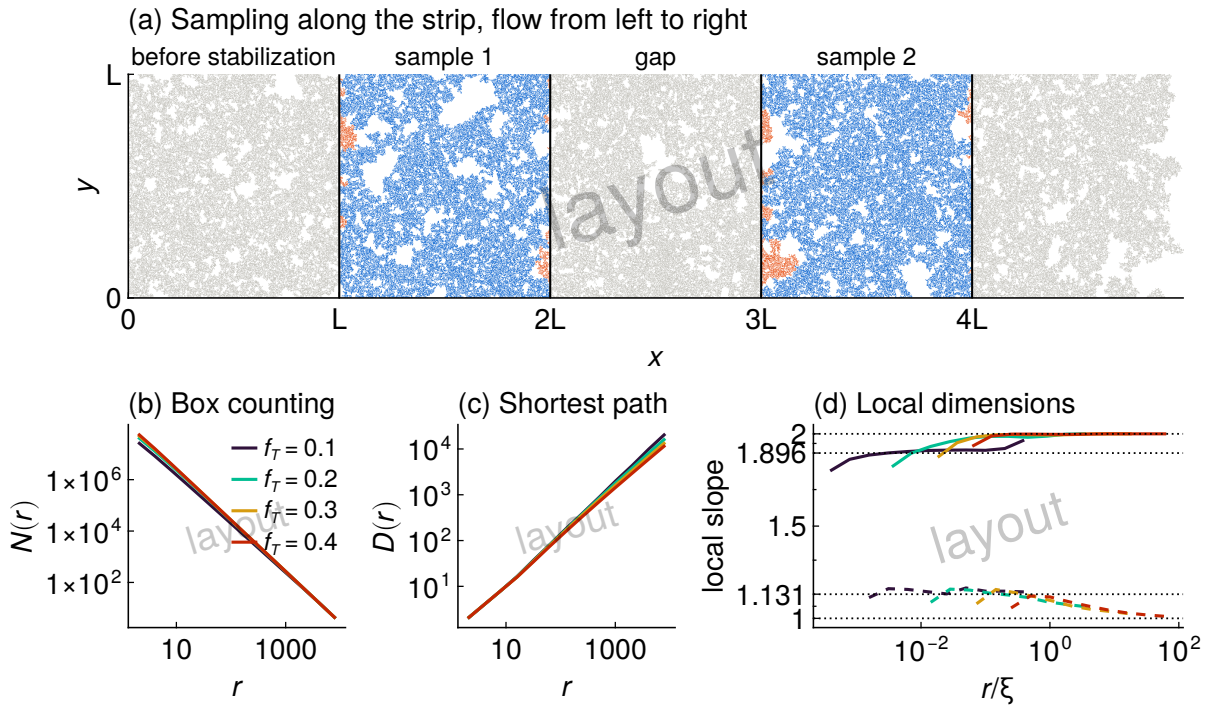


Figure 5: Fractal dimensions at fixed rate. (a) Sampling along a strip of one run, with $L = 2^{10}$, $f_T = 0.2$ and flow from left to right. In each sample, blue is the largest cluster that spans the window along the flow, and orange the other clusters. Regions outside the samples are in gray. (b–d) Site percolation with $L = 2^{14}$, $c = 0.05$, $\rho_0 = 0.2$, random seeds and one window of side L per run, for 12 runs: (b) box counts, (c) shortest paths, and (d) local slopes of $N(r)$ (solid) and $D(r)$ (dashed), over one factor of 4 in r, against r/ξ, with $\xi = (p^* - p_c)^{-4/3}$. Dotted lines mark 1, $d_{\min}$, d_f and 2.

Uses: Fig. 5 sets the layout only. Germano and Junior will produce the data with their code.
Missing: Final data for panels (b–d); the numbers in the text of this section come from a first pass with 12 windows and should be updated.

7. Limits of the one-chance model

The feedback rule rests on the volume balance alone, and the meaning it gives to f_T and c does not depend on the growth rule. The growth rule is a different matter. A pore that fails its test in SOP stays empty forever, whereas a real pore keeps its threshold and can be entered later if the pressure rises. The one-chance rule is therefore part of the model we started from, not a consequence of the injection problem.

The rates are the second limit. The results above hold for $f_T \geq 0.08$, which through Eq. (12) corresponds to capillary numbers of order 0.1, where the viscous forces neglected in the derivation are no longer small. Slow drainage lies several orders of magnitude below, as the estimates in Section S5 of the Supplemental Material show, and the one-chance front cannot follow it there, since it dies out at small f_T. At such rates the imposed flow can also fill less than one pore per step, so the synchronous front of the model no longer applies.

SOP is therefore not a model of slow drainage. What carries over is the structure of the problem: a shared pressure opens pathways, the pathways relax the pressure, and the volume balance fixes how the two are coupled. A description of real flow needs a growth rule in which pores keep their thresholds, which we leave for future work.

8. Conclusions

We have shown that the feedback of self-organized percolation is a volume balance, and that this reading explains why the model is critical. A compressible system injected at a constant rate stores the volume that the pores do not take up, and the pressure it builds is the occupation probability of the medium. The target of the feedback is then the imposed flow, and its strength is the stiffness of the system, which gives measurable meaning to the two parameters of the model.

The main result is the origin of the plateau. At a fixed p, the growth front has a stationary density $f_\infty(p)$ that vanishes at p_c, and the feedback only finds the value at which this density equals the target, $p^* = f_\infty^{-1}(f_T)$. The system with feedback and the system without it agree for $f_T \geq 0.2$, in site and bond percolation. Self-organization here does not create a new state: it selects an existing one, and a thin front can only exist near the threshold. The form of f_∞ follows from the geometry of the cluster, with an exponent $x = \beta + \nu(d_{\min} - 1) \approx 0.313$ and no fitted parameters, and it gives a pressure–rate curve, $p^* - p_c \sim f_T^{1/x}$, which we measured at fixed size.

Three consequences follow. The stationary state is robust: it does not depend on the initial value of p, on the stiffness at moderate feedback or, beyond a boundary layer about one gap thick, on the layout of the seeds. The threshold can be located from plateaus at a sequence of rates and sizes, with no tuning of the rate to the size, to about 3×10^{-4}. And the rate sets the distance from criticality, and with it a correlation length: a slow injection leaves a fractal cluster over a wide range of scales, and a fast one is fractal only at short distances.

Two points remain open. Below $f_T \approx 0.15$ the plateau departs from the fixed-p curve, by an amount that shrinks with size, and finite-size effects, slow relaxation and the fluctuations of p have not been separated. More importantly, the one-chance rule is a property of the model and not of fluids, so SOP does not describe slow drainage. The next step is therefore a growth rule in which pores keep their thresholds, within the same volume balance.

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